

Linearly Stable Self-Similar Expansion of Four and Five Point Vortices

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Abstract

Point vortices can move self-similarly, keeping their shape while the configuration grows like $\sqrt{t}$ and turns. For three vortices such expanding configurations are stable, and Zbarsky used this to show that vortex patches placed at the three vortices stay confined for all time; he wrote that the result would most likely carry over to stably growing systems of four or more vortices, given sufficiently good stability. We give two such configurations that are linearly stable modulo their symmetries. For the circulations $(-1, -5/2, -1/9, 4/5)$ and $(-1, 3/7, 7/8, -9/7, -47/35)$ there are self-similarly expanding configurations of four and five Euler point vortices whose linearization in similarity variables has, besides the double eigenvalue 0 of the rotation and of the one-parameter family of such configurations, only eigenvalues with real part -1 or -2 : in the invariant complement of the rotation and the family, every solution of the linearized equation decays, relative to the size of the configuration, at least like the inverse of the size. The proof is computer-assisted: the configurations are enclosed by the Krawczyk method in ball arithmetic, and the spectrum is controlled through an exact lemma, which shows that six eigenvalues are forced by the rotation, the scaling, the translations and the family (one of them through a pairing) and that the others come in pairs $k, 2 - k$, together with enclosures of traces of powers of the Jacobian. The property is open, so such configurations form open sets; in a random sample about one four-vortex collapse in seven and one five-vortex collapse in seventeen reverses into a linearly stable expansion. We do not address whether this linear stability suffices for Zbarsky's confinement argument.

Keywords: point vortices; self-similar motion; linear stability; computer-assisted proof; interval arithmetic.

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1 Introduction

N point vortices with circulations Γ_j at positions $z_j \in \mathbb{C}$ move by

$$\dot{\bar{z}}_j = \frac{1}{2\pi i} \sum_{k \neq j} \frac{\Gamma_k}{z_j - z_k}. \quad (1)$$

A motion is self-similar when the configuration keeps its shape while it grows or shrinks and turns; this happens when $\dot{z}_j = \kappa(z_j - z_c)$ for all j at $t = 0$, with one $\kappa \in \mathbb{C}$, where $z_c = \sum_j \Gamma_j z_j / \sum_j \Gamma_j$ [4, Sect. 2]. Then the size of the configuration grows or shrinks like $|1 + 2t \operatorname{Re} \kappa|^{1/2}$, the configuration

expands when $\text{Re } \kappa > 0$ and collapses in finite time when $\text{Re } \kappa < 0$ [7, 9, 1, 2, 6]. Reversing every circulation reverses time, so each collapse becomes an expansion of the same shape, and conversely.

Zbarsky [11] proved that for a generic self-similarly expanding configuration of three vortices one may replace each vortex by a small patch of vorticity and the patches stay confined for all time, their radius growing more slowly than their distance. The restriction to three vortices comes from the stability analysis. He writes that “if one found stably growing systems of 4 or more vortices, the confinement result would most likely carry over with little modification. However, one needs sufficiently good stability results; orbital linear stability, which may not be hard to obtain for some systems, is not enough” [11, pp. 4–5 of the arXiv version]; the same point recurs in [12, after Cor. 16]. For three vortices, stability of the expanding family goes back to Tavantzis and Ting [10]; Iwayama and Yajima studied the linear stability of the self-similar motions of three vortices in the α -models [5].

We show that linearly stable expansions of four and five vortices exist. Section 2 sets up similarity variables, in which a self-similar motion is a fixed point, and proves a lemma on the structure of the spectrum of the linearization (Lemma 1): six eigenvalues are forced by the rotation, the scaling, the translations and the one-parameter family of self-similar configurations with the given circulations (the second eigenvalue 2 through a pairing), and the others come in pairs $k, 2 - k$. For three vortices nothing else is left. For four vortices one pair is left, and a single number decides its stability; for five, two pairs. Section 3 states and proves, with a computer-assisted proof, that there are configurations of four and of five vortices whose reversal expands with every other exponent of real part 1 (Theorems 1 and 2, Corollary 1). Section 4 gives numerical illustrations, controls and a sample of how common such configurations are, and Section 5 discusses what is and is not shown.

2 Similarity variables and the structure of the spectrum

2.1 Similarity variables

Let $V(\zeta) = (V_1, \dots, V_N)$ be the right side of (1), $V_j = (i/2\pi) \sum_{k \neq j} \Gamma_k / (\zeta_j - \zeta_k)$. It is homogeneous of degree -1 , $V(\lambda\zeta) = \lambda^{-1}V(\zeta)$ for real $\lambda > 0$, and equivariant under rotations, $V(e^{i\phi}\zeta) = e^{i\phi}V(\zeta)$. For $\kappa = a + i\omega$ put $z_j(t) = \rho(t)e^{i\omega\tau}\zeta_j(\tau)$ with $\rho^2 = 1 + 2at$ and $d\tau/dt = \rho^{-2}$; then (1) becomes

$$\frac{d\zeta}{d\tau} = V(\zeta) - \kappa\zeta, \quad (2)$$

and a self-similar motion with this κ and $z_c = 0$ is a fixed point of (2). We fix the scale of time by $2\pi a = -1$ for a collapse, write $2\pi\kappa = -1 + ib$, and work with $E(\zeta) = 2\pi(V(\zeta) - \kappa\zeta) = 2\pi V(\zeta) + (1 - ib)\zeta$ and the rescaled time $s = \tau/2\pi$; for a collapse, $s = -\ln \rho$. The winding number of the collapse is $P = |b|/2$ [4]. Let DE be the real $2N \times 2N$ Jacobian of E at a zero ζ^* . A perturbation along an eigenvector with eigenvalue k behaves like $e^{ks} = \rho^{-k}$ relative to the size ρ of the configuration. Writing $\delta\zeta = u + iv$, one finds

$$DE = \begin{pmatrix} A_r + I & A_i + bI \\ A_i - bI & -A_r + I \end{pmatrix}, \quad A_{jk} = \frac{i\Gamma_k}{(\zeta_j - \zeta_k)^2} \ (k \neq j), \quad A_{jj} = -\sum_{k \neq j} A_{jk}, \quad (3)$$

with $A = A_r + iA_i$ evaluated at ζ^* .

Time reversal. If ζ^* is a collapse for the circulations Γ , it is a fixed point of the similarity dynamics of the circulations $-\Gamma$ with κ replaced by $-\kappa$, an expansion with the same winding, and the Jacobian of its similarity dynamics is $-DE$; the similarity time is then $s = \ln \rho$. So a mode with eigenvalue k of DE behaves in the expansion like ρ^{-k} relative to its size, and it decays as the configuration grows exactly when $\text{Re } k > 0$.

2.2 The structure of the spectrum

Lemma 1. *Let ζ^* be a zero of E , with all $\Gamma_j \neq 0$ and the ζ_j^* distinct, and suppose that near (ζ^*, b) the zeros (ζ, b) of E , with the circulations fixed, form up to rotation a smooth curve through (ζ^*, b) whose ζ -component is transverse to the rotations. Then the characteristic polynomial p of DE satisfies $p(\lambda) = p(2 - \lambda)$ and is divisible by*

$$\lambda^2(\lambda - 2)^2((\lambda - 1)^2 + b^2),$$

and the quotient q , of degree $2N - 6$, also satisfies $q(\lambda) = q(2 - \lambda)$. In particular the $2N - 6$ remaining eigenvalues, counted with multiplicity, come in pairs $k, 2 - k$.

Proof. Rotation. $E(e^{i\phi}\zeta) = e^{i\phi}E(\zeta)$, so differentiating at $\phi = 0$ gives $DE(i\zeta^*) = iE(\zeta^*) = 0$.

Scaling. $E(\lambda\zeta) = 2\pi V(\zeta)/\lambda + (1 - ib)\lambda\zeta$; at $\lambda = 1$ the derivative is $-2\pi V(\zeta^*) + (1 - ib)\zeta^* = 2(1 - ib)\zeta^*$, since $2\pi V(\zeta^*) = -(1 - ib)\zeta^*$. So $DE\zeta^* = 2\zeta^* - 2bi\zeta^*$, and on the span of ζ^* and $i\zeta^*$ the map DE is triangular with eigenvalues 2 and 0.

Translations. V is invariant under $\zeta \mapsto \zeta + c(1, \dots, 1)$, so DE acts on $\mathbf{1} = (1, \dots, 1)$ and $i\mathbf{1}$ as multiplication by $1 - ib$: $DE\mathbf{1} = \mathbf{1} - bi\mathbf{1}$, $DE(i\mathbf{1}) = i\mathbf{1} + b\mathbf{1}$, with eigenvalues $1 \pm ib$.

The family. Along the curve of zeros, $E(\zeta(\sigma); b(\sigma)) = 0$, and $\partial_b E = -i\zeta$. Differentiating, $DE\zeta'(0) = b'(0)i\zeta^*$, so $DE^2\zeta'(0) = 0$. As $\zeta'(0)$ is not a real multiple of $i\zeta^*$, the generalized kernel of DE has dimension at least 2: the algebraic multiplicity of 0 is at least 2.

Pairing. With $H = -\frac{1}{2}\sum_{j \neq k} \Gamma_j \Gamma_k \ln |\zeta_j - \zeta_k|$ and $I = \sum_j \Gamma_j |\zeta_j|^2$, the field $\zeta \mapsto 2\pi V(\zeta) - ib\zeta$ is the Hamiltonian vector field of $H + (b/2)I$ for the symplectic form $\sum_j \Gamma_j dx_j \wedge dy_j$, which is nondegenerate because every $\Gamma_j \neq 0$ (Kirchhoff's form of (1) and the rotations generated by I , up to the sign convention of the form). The linearization of a Hamiltonian vector field at any point is an infinitesimally symplectic matrix, whose characteristic polynomial is even. Since DE is this linearization plus the identity, $p(\lambda) = r(\lambda - 1)$ with r even, that is $p(\lambda) = p(2 - \lambda)$, with multiplicities.

Conclusion. By pairing, the multiplicity of 2 equals that of 0, at least 2, so $\lambda^2(\lambda - 2)^2$ divides p . The span of $\mathbf{1}$ and $i\mathbf{1}$ is invariant, and the characteristic polynomial of DE on it, $(\lambda - 1)^2 + b^2$, divides p too; its roots $1 \pm ib$ are never 0 or 2, so the two factors are coprime and p is divisible by the displayed product. That product is invariant under $\lambda \mapsto 2 - \lambda$, hence so is the quotient. $\square$

For three vortices $2N - 6 = 0$: in similarity variables every exponent of a three-vortex expansion is forced by the symmetries and the family, which is the linear shadow of the classical stability of the expanding three-vortex family [10]. For four vortices one pair $k, 2 - k$ remains, and with $c = k(2 - k)$ the sum of squares of all eigenvalues gives

$$\text{tr}(DE^2) = (0 + 0 + 4 + 4 + 2(1 - b^2)) + (4 - 2c), \quad c = 3 + 3b^2 - \text{tr}(A\bar{A}), \quad (4)$$

using $\text{tr}(DE^2) = 2\text{tr}(A\bar{A}) + 2N(1 - b^2)$, which follows from (3). Both remaining exponents have positive real part exactly when $c > 0$; when $c > 1$ they are $1 \pm i\sqrt{c - 1}$. For five vortices two pairs remain; with $u = (k - 1)^2$ they are the roots u_1, u_2 of $u^2 - su + p$, where $s = u_1 + u_2$ and $u_1^2 + u_2^2$ follow from $\text{tr}((DE - I)^2)$ and $\text{tr}((DE - I)^4)$ after the contributions $4 - 2b^2$ and $4 + 2b^4$ of the six forced eigenvalues are removed. Every remaining exponent has positive real part exactly when every root u satisfies $|\text{Re} \sqrt{u}| < 1$; in particular it suffices that both roots are real and less than 1.

3 Results

Theorem 1 (Four vortices). *For the circulations $\Gamma = (1, 5/2, 1/9, -4/5)$ there is a self-similar collapse of four Euler point vortices with $2\pi\kappa = -1 + ib$, winding*

$$P = b/2 = 2.5258464741344909107270\dots,$$

and z_4 with real part $8/25$ in the gauge $z_1 \in \mathbb{R}$, $z_c = 0$, whose Jacobian DE has the six eigenvalues $0, 0, 2, 2, 1 \pm ib$ of Lemma 1 and the pair $1 \pm i\omega$ with $\omega = 4.4588620962481\dots$; here $c = 20.881451193358\dots$ in (4).

Theorem 2 (Five vortices). *For the circulations $\Gamma = (1, -3/7, -7/8, 9/7, 47/35)$ there is a self-similar collapse of five Euler point vortices with winding $P = 2.40234703541713841\dots$ and z_2 with real part $3/5$ in the same gauge, whose Jacobian DE has the six eigenvalues of Lemma 1 and the four eigenvalues $1 \pm 3.4538555\dots i$ and $1 \pm 3.1885675\dots i$.*

Corollary 1. *For the circulations $(-1, -5/2, -1/9, 4/5)$ and $(-1, 3/7, 7/8, -9/7, -47/35)$ the configurations of Theorems 1 and 2 expand self-similarly, growing like $\sqrt{t}$. In similarity variables the linearization at them has the double eigenvalue 0, whose generalized eigenspace consists of the rotation and the tangent of the one-parameter family of expanding configurations with the same circulations, and every other eigenvalue has real part -1 or -2 . In the invariant complement of the rotation and the family, every solution of the linearized equation therefore decays, relative to the size ρ of the configuration, at least like ρ^{-1} , that is like $t^{-1/2}$.*

Proof of the theorems. The program `verify_stable_expansion.py` (Section 4) works in the gauge of [4, Sect. 8]: $\Gamma_1 = 1$, z_1 real, $\sum_k \Gamma_k/(z_j - z_k) = \lambda \bar{z}_j$ with $\lambda = 2P - i$, which is $E = 0$ with $b = 2P$ and $z_c = 0$. For four vortices it fixes $x_4 = 8/25$, $\Gamma_2 = 5/2$ and $\Gamma_3 = 1/9$ and proves by the Krawczyk test, in the ball arithmetic of FLINT/Arb, that the 8 equations have exactly one zero in a box of radius 10^{-4} around the numerical solution; on the box the circulations are nonzero, the total circulation is nonzero, the vortices are distinct and away from the origin, and the necessary conditions $\sum_j \Gamma_j z_j = 0$, $\sum_{j < k} \Gamma_j \Gamma_k = 0$ and $\sum_j \Gamma_j |z_j|^2 = 0$ are enclosed with 0. The second of these forces $\Gamma_4 = -4/5$. The Jacobian of the square system is invertible on the box, so by the implicit function theorem the zeros with Γ_2, Γ_3 fixed form a smooth curve parametrized by x_4 ; along it Γ_4 stays $-4/5$, and its tangent v has $\text{Im } v_1 = 0$ and v_4 with real part 1, while $i\zeta^*$ has $\text{Im}(i\zeta^*) = x_1 \neq 0$, so v is not a real multiple of $i\zeta^*$. The hypotheses of Lemma 1 hold, and the program also checks on the box that the vectors of its proof satisfy their identities. It then encloses $c = 3 + 3b^2 - \text{tr}(A\bar{A})$ by $20.8814511933587 \pm 1.4 \cdot 10^{-14}$, so $c > 1$ and the remaining pair is $1 \pm i\sqrt{c-1}$; a second enclosure from $\text{tr}((DE - I)^2)$ agrees. For five vortices it fixes $x_2 = 3/5$ and $\Gamma_2, \Gamma_3, \Gamma_4$, proves existence the same way, finds $\Gamma_5 = 47/35$, checks the hypotheses of the lemma, and encloses s and p ; the discriminant $s^2 - 4p$ is positive and both roots, $u_1 = -11.929118\dots$ and $u_2 = -10.166963\dots$, are negative, so the four remaining eigenvalues are $1 \pm i\sqrt{-u_1}$ and $1 \pm i\sqrt{-u_2}$. $\square$

Proof of the corollary. By the time reversal of Section 2 the configurations expand for the negated circulations, with Jacobian $-DE$ in similarity variables and similarity time $s = \ln \rho$. The eigenvalues of $-DE$ are $0, 0, -2, -2, -1 \mp ib$ and, by the theorems, $-1 \mp i\omega$ or $-1 \mp i\sqrt{-u_{1,2}}$. By the theorems no remaining eigenvalue is 0, so the generalized eigenspace of 0 is exactly two-dimensional and, by the proof of Lemma 1, spanned by $i\zeta^*$ and the tangent of the family. The eigenvalues with real part -1 are simple: $\omega \neq b$ for four vortices and $\sqrt{-u_1}, \sqrt{-u_2}$ and b are distinct for five, which the program checks. The eigenvalue -2 is double, the partner of the one at 0, and even as a Jordan block it contributes only $s e^{-2s}$. On the complementary invariant subspace every solution of the linearized equation therefore decays at least like $e^{-s} = \rho^{-1}$, and ρ^2 grows linearly in t . $\square$

program	what it checks	checks
<code>verify_stable_expansion.py</code>	Theorems 1, 2, the hypotheses of Lemma 1 on the boxes, controls, integrations	50
<code>survey_expansions.py</code>	the random sample of Section 4 (numerical)	

Table 1: The programs. Each prints every check and stops with an error if one fails; the output is in `data/`.

Remark 1 (Open sets). The conclusions are open conditions: c , and the roots u , depend continuously on the configuration and the circulations along the collapse set, which near these points is a smooth manifold of dimension $N - 1$ (modulo translations, rotations and the scalings of lengths and circulations) [4, Sect. 7]. So every collapse near those of the theorems, with nearby circulations, reverses into a linearly stable expansion as well; the stable expansions of four and five vortices form open subsets of the $(N - 1)$ -dimensional collapse set, not isolated examples.

4 Numerical illustration, controls and a sample

Controls. The same program certifies two controls, with the same side conditions. A four-vortex collapse with circulations $(1, -1003/1000, 13/10, \Gamma_4)$ has $c = -47.7346 \dots < 0$: its remaining pair is real, $k < 0$, and its reversal is unstable. A three-vortex collapse has $\text{tr}((DE - I)^2) = 4 - 2b^2$ to within its enclosure, the contribution of the six forced eigenvalues alone, as Lemma 1 requires for $N = 3$.

Direct integration (binary64, DOP853, relative tolerance 10^{-12} ; not part of the proof). The four-vortex expansion of Corollary 1, started from the midpoint of the certified box, keeps its shape to $4 \cdot 10^{-15}$ while its size grows 178-fold; a random perturbation of 10^{-5} in each coordinate (a shape deviation of $3.3 \cdot 10^{-5}$) stays below $2 \cdot 10^{-4}$, settling on a nearby member of the family, the neutral direction. The reversal of the unstable control, perturbed by 10^{-8} in each coordinate, departs from its shape by $1.5 \cdot 10^{-3}$ when its size has grown eightfold and by 0.3 at sixteenfold. A self-similar collapse, by contrast, departs from self-similarity in binary64 under round-off alone [8, 3].

How common. A least-squares solver from random starts, in binary64 with fixed seeds, gave 342 converged four-vortex collapses (not checked for duplicates), of which 52 have $c > 0$, and 543 five-vortex collapses, of which 32 have every remaining exponent with positive real part (`survey_expansions.py`). This is a sample of what a naive search finds, not a measure on the collapse set. Of the 52 four-vortex cases, 37 have the remaining pair on $\text{Re } k = 1$, as in Theorem 1, and 15 a real pair $0 < k < 2$. Of the 32 five-vortex cases, 10 have both remaining pairs on $\text{Re } k = 1$, as in Theorem 2, 16 a quadruple $1 \pm x \pm iy$ with $0 < x < 1$, and 6 a real pair $0 < k < 2$.

5 Discussion

The theorems give self-similarly expanding configurations of four and five vortices that are linearly asymptotically stable modulo the rotation and the family of expansions with the same circulations, with a spectral gap: in the invariant complement of those directions every solution of the linearized equation decays, relative to the size, like a negative power of t . This is a linear statement. Zbarsky writes that orbital linear stability, “which may not be hard to obtain for some systems, is not enough” [11]; we have not examined whether the spectral gap here is enough for his argument, or whether his confinement argument [11, 12] goes through for these configurations, and we have not proved nonlinear stability, although a spectral gap of this kind is the usual starting point for it.

The circulations of the theorems are not symmetric and the configurations have no symmetry; by Remark 1 they are not special.

The examples are Euler vortices. The α -models, where a vortex induces the velocity $\Gamma r^{-\alpha-1}/(2\pi)$, have the same structure after the obvious changes of homogeneity, and we have not treated them. Nor have we asked which fraction of the collapse set reverses into stable expansions, or whether stable expansions exist for every $N \geq 4$; the sample of Section 4 suggests that they are common for $N = 4$ and 5.

Data availability. The programs `verify_stable_expansion.py` and `survey_expansions.py`, with the modules `certify_ball_ad.py` and `certify_pipeline.py` of [4], their inputs and their output will be in the repository <https://github.com/ChaseHendrick/stable-expansion>: the programs in `code/` and their inputs and output in `data/`.

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